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|    |   |                                                                                                                                                                                                                                                                                                                                                           |
| 29 | D | <p>Original activity is <math>180 \text{ counts s}^{-1}</math> after deducting background count. After 10 hours, the radioactive sample decays for 2 half-lives.</p> <p>Hence <math>\frac{1}{4}</math> of its original activity remains giving <math>45 \text{ counts s}^{-1}</math>. Adding background counts giving <math>75 \text{ s}^{-1}</math>.</p> |
| 30 | A | $\frac{1}{111} \times \frac{1}{111} \times 0$                                                                                                                                                                                                                                                                                                             |